Janghyun **Noh**

AI Researcher · Bioinformatics

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Research Interest

Bioinformatics, Medical AI, Deep Learning, Multimodal Learning, Single-cell RNA Sequencing Analysis

Education

Myongji University — *Seoul, S.Korea*

M.S. in Artificial Intelligence — *Sep. 2024 - Aug. 2026*

- Conducted research at the Artificial Intelligence and Data Analytics Laboratory under the supervision of Prof. Minsik Oh.
- Conducted collaborative research with external institutions including the Laboratory of Immune Tolerance at Seoul National University Hospital (SNUH), the Korea Disease Control and Prevention Agency (KDCA), and Kyungpook National University.
- Served as a teaching assistant for undergraduate courses including Deep Learning, Basic Programming, and Data Mining.

Myongji University — *Seoul, S.Korea*

B.S. in Convergence Software and Data Technology — *Mar. 2018 - Aug. 2024*

- Conducted undergraduate research at the Artificial Intelligence and Data Analytics Laboratory under the supervision of Prof. Minsik Oh.
- Conducted collaborative research on clinical and genetic factors associated with hypertension progression in patients with hyperlipidemia in collaboration with the Korea Disease Control and Prevention Agency (KDCA).

RED School – Alternative Middle-High School — *Geumsan, S.Korea*

Student — *Mar. 2012 - Jan. 2018*

- Received academic excellence scholarships for two consecutive years.
- Participated in student leadership and community activities.

Research Experience

Artificial Intelligence and Data Analytics Lab, Myongji University — *Seoul, S.Korea*

Undergraduate Research Intern / Graduate Researcher — *Dec. 2022 - Present*

- Developed scCap, an annotation-free phenotype prediction framework for single-cell RNA-seq data.
- Conducted differential expression and spatial transcriptomics analyses for biomarker discovery in cirrhotic and non-cirrhotic HCC tissues.
- Conducted spatial transcriptomics analyses of TNBC Xenium datasets for macrophage subtype validation and tumor microenvironment characterization.
- Developed COVID-19 severity prediction models using multimodal clinical and imaging data.
- Developed multimodal medical AI models using electronic health records, chest X-ray images, and clinical notes from MIMIC datasets.
- Implemented bioinformatics pipelines for variant discovery and candidate gene identification in apple whole-genome sequencing data.
- Conducted clinical and genomic analyses of KoGES cohort data for hypertension risk stratification.

Publications & Presentations

Journal Articles

Annotation-free phenotype prediction using knowledge-augmented clustering from single-cell RNA sequencing data — *Briefings in Bioinformatics (BIB)*

Accepted — *2026*

J. Noh, Y. Shin, M. Kim, and M. Oh

Ferroptosis-based therapeutic strategy targeting iron metabolism and SLC7A11 overcomes oxaliplatin resistance in patient-derived pancreatic cancer organoids — *Journal of Experimental & Clinical Cancer Research (JECCR)*

Published — *2026*

J. Choi, K. Lee, M. Oh, **J. Noh**, Y. Kim, E. Hwang, I. Cho, W. Paik, J. Ryu, J. Ku, and S. Lee

Conference Presentations

Patient-Level Metacell Construction with Pre-trained Embeddings for Phenotype Prediction from scRNA-seq — *KIISE Korea Computer Congress (KCC)*

Oral Presentation Accepted — *2026*

Y. Shin, **J. Noh**, and M. Oh

Lightweight Analysis of Patient Phenotype Prediction Based on Single-Cell RNA-seq Data Reduction Techniques
KIISE Korea Software Congress (KSC)
Poster Accepted (Best Poster Award) — *2025*
Y. Shin, **J. Noh**, and M. Oh

An annotation-free framework for phenotype prediction using hierarchical multiple instance learning with single-cell RNA-sequencing data
International Conference on Genome Informatics ISCB-Asia
Poster Accepted — *2025*
J. Noh, Y. Shin, M. Kim, and M. Oh

Patient Phenotype Prediction from Single-cell RNA-seq Data through Annotation-free Multiple Instance Learning
BIOINFO 2025, Korean Society for Bioinformatics (KSBI)
Poster Presentation (Excellent Poster Award) — *2025*
J. Noh, Y. Shin, M. Kim, and M. Oh

AMIL: Automated Cell Grouping Framework for Phenotype Prediction with Multiple Instance Learning
KIISE Korea Computer Congress (KCC)
Oral Presentation (Best Presentation Award) — *2025*
J. Noh, M. Kim, and M. Oh

Predicting In-Hospital Mortality in Patients with Acute Respiratory Distress Syndrome Using MIMIC-III Data: A Comparative Study of Machine Learning Models with Varied Missing Data Handling Strategies
KIISE Korea Software Congress (KSC)
Poster Presentation — *2023*
J. Noh, G. Choi, and M. Oh

Honors & Awards

Awards

2025	**Best Poster Award**, KIISE Korea Software Congress (KSC)	*Seoul, S.Korea*
2025	**Excellent Poster Award**, BIOINFO 2025, Korean Society for Bioinformatics (KSBI)	*Seoul, S.Korea*
2025	**Best Presentation Award**, KIISE Korea Computer Congress (KCC)	*Seoul, S.Korea*
2022	**President's Merit Award**, Korea Trade-Investment Promotion Agency (KOTRA)	*Seoul, S.Korea*

Scholarships

2022	**TriBIG Project Scholarship**, Korea Trade-Investment Promotion Agency (KOTRA)	*Seoul, S.Korea*
2019	**Globalization Scholarship**, Myongji University	*Seoul, S.Korea*
2018	**Scholarship III**, Myongji University	*Seoul, S.Korea*
2016	**Academic Excellence Scholarship**, RED School	*Geumsan, S.Korea*
2015	**Academic Excellence Scholarship**, RED School	*Geumsan, S.Korea*

Projects

2025 – 2026 CRISPR-Cas9 유도 대규모 DNA 결실 검출·정량화를 위한 클라이언트 사이드 고속·고정밀 웹 기반 분석 플랫폼 개발
2025 – 2026 HCC GeoMx 공간전사체 기반 간경변·비간경변 조직의 병리적 차이 규명 및 면역조절 타겟 발굴 연구
2025 – 2025 멀티모달 임상·영상 데이터를 활용한 코로나19 환자 중증도 예측 모델·환경 구축
2024 – 2029 기후 변화 대응을 위한 사과꽃 휴면타파 및 꽃 발달 시기의 유전적 생리학적 메커니즘 연구
2024 – 2029 공간 전사체 기반 타겟 발굴과 in silico 및 오가노이드 활용 타겟 검증을 통한 난치성 고형암 MLT 신규 치료제 신속 개발
2024 – 2028 AI 기반 타겟 발굴 플랫폼 확립을 통한 만성 염증성 질환 조절 CAR-Treg 세포 치료제 개발
2023 – 2024 KoGES 코호트 기반 고지혈증 환자의 고혈압 진행 위험군 규명을 위한 임상·유전적 차이 분석 연구

Technical Skills

Programming	Python, R
Frameworks	PyTorch, TensorFlow, Scikit-learn, Scanpy
Tools	Linux, Git
Research	Medical AI, Bioinformatics, Deep Learning, Multimodal Learning, Single-cell RNA Sequencing Analysis
Languages	Korean, English

Extracurricular Activities

Domestic

Google Developer Student Clubs, Myongji University — *Seoul, S.Korea*

AI Team Member — *Oct. 2022 - Dec. 2023*

- Engaged in machine learning and deep learning study groups
- Collaborated on AI projects and participated in AI competitions

Big Leader AI Academy — *Seoul, S.Korea*

Project Member — *Jul. 2022 - Oct. 2022*

- Developed a customs notice analysis and monitoring system in collaboration with the Big Data Team of KOTRA
- Participated in an industry-collaborative AI project for KOTRA internal service applications (TriBIG)
- Received a scholarship through outstanding performance in the TriBIG project

Data Youth Campus, Dongguk University — *Seoul, S.Korea*

Participant — *Jul. 2022 - Aug. 2022*

- Completed a data science-based intelligent software program affiliated with Korea Data Agency (KDATA)
- Conducted practical projects and training in machine learning and data analysis

International

Medical Mission Trip to the Philippines (GMMA) — *Manila, Philippines*

Volunteer — *Jan. 2024*

- Participated in medical volunteer activities with international healthcare professionals
- Provided community support and engaged in intercultural collaboration activities

International Work Camp in Iceland — *Reykjavik, Iceland*

Volunteer — *Dec. 2019 - Jan. 2020*

- Participated in environmental campaigns and community workshops with local residents
- Collaborated with international volunteers through intercultural exchange activities

Mission Trip to Vietnam — *Hanoi, Vietnam*

Volunteer — *Aug. 2018*

- Participated in intercultural exchange activities with local residents and university students
- Contributed to local community and volunteer activities

Others

Language

2023 **TOEIC**, Total 920 (RC 430, LC 490)

Certifications

2023 **Engineer Information Processing**, Human Resources Development Service of Korea

2023 **SQL Developer (SQLD)**, Korea Data Industry Promotion Agency